Analog and Digital Design: World Trivia Game

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Abstract

World Trivia is an interactive trivia board game designed to promote mental stimulation and social engagement in older adults or anyone with mobility limitations. To support users with visual limitations, we incorporated a large LCD display for clear text and graphics, a palm-sized joystick for easy navigation, and bright LEDs to guide players.

An Arduino microcontroller serves as the core of the system, coordinating user input, visual output, and game logic. This main point of connection helps to keep our design focused on the goal of accessibility within our scope. Throughout the development process, we prioritized ease of use, minimizing the number of required inputs and providing clear visual feedback. By focusing on accessibility, clarity, and enjoyment, World Trivia illustrates that assistive and inclusive technologies do not have to feel clinical or intimidating; instead, they can be playful, engaging, and empowering.

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1 Introduction

1.1 Project Overview

World Trivia was designed for four players each with personal LCD screen displays. Their screens are mounted on the board below personal input buttons for answering A through D. As the center point of the board, World Trivia has a laser engraved map of the world, with an LED at the center of each continent. Below this map is a joystick to navigate to one of the seven continents, engraved with a compass rose. At the top of the board is a larger LCD screen presenting the trivia questions.

1.2 Inspiration/Client

Our project was designed with the elderly in mind. We liked the idea of creating a product that could keep elderly people's minds engaged while bringing people together. Users would be able to reminisce over past travel or dream of future travel.

2 Project Scope

The man goal for our project was to make and design a trivia game that is mentally stimulating. Our ideal audience were people with mobility limitations that they could use our product without difficulty.

2.1 Goals

Our goals were to learn new skills and create our project. We wanted to use the Arduino GIGA to make a stable system, to learn how to display questions and answers using the GIGA LCD, to learn how to map joystick movement to the seven continents, and choose one using the joystick, track each players score, to keep our game powered and running, and wanted to put it all together to create a playable game.

2.2 Plans to Achieve Goals

We planned to purchase some hardware components and custom-build some others:

We purchased an Arduino GIGA, a shield for GIGA Arduino, 4 LCD screens, string LEDs, a joystick, an Acrylic sheet, stock wood, and lithium-ion batteries. We had access to buttons, wires, and solder as well. We knew we wanted to build some components both for aesthetics and functionality. The game shell was built with dimensions specific to our electrical components and to best suit our intended audience of the elderly. We built a joystick with more ergonomic dimensions, with a compass rose design on top to match the world map theme.

2.3 Team Assignments

Daniela Chure was assigned to physical game construction. She modeled and refined the external housing, the placement of LEDs, and the layout dimensions, in addition to soldering.

Sarah Takken was in charge of structural fabrication. She also constructed the game board housing, and created the laser engraving, the joystick CAD, and print.

Madison Dewrell was in charge of software and game logic. She created game code and logic, including joystick mapping, trivia questions, timing, scoring, and LCD output.

Adelyn Buss was in charge of the electrical design. She made schematics, laid out wiring, put together electrical components, tested components, soldered, and assisted with coding.

3 Specifications

The specifications for World Trivia establish the required electrical, mechanical, and user-interface performance parameters that guide all design decisions. These specifications ensure that the system remains safe, accessible, and reliable during repeated gameplay. The electrical specifications—including voltage ranges, current limits, and screen sizes—were chosen to maintain compatibility with the Arduino GIGA and all peripheral devices. Mechanical specifications such as board dimensions and component spacing were selected to maximize comfort and usability for older adults, providing clear visual separation between inputs and ensuring that users with limited dexterity can interact with the system easily. Timing and gameplay specifications were developed to balance challenge and accessibility by giving players adequate time to read questions, navigate with the joystick, and select responses.

Table 1: Quantitative and Qualitative Specifications

Specification	**Minimum**	**Nominal**	**Maximum**	**Units**
Operating Voltage	4.8	5.0	5.2	V
Total Current Draw	–	750	1000	mA
Large LCD Screen Size	–	3.5	–	in
Player LCD Count	4	4	4	–
Joystick Directions	4	7 mapped positions	7	–
Question Time Window	5	6	10	seconds
Game Board Dimensions	23	24	25	in (width)
LED Count	7	7	7	–
Power Supply Rating	–	7.4	–	V Li-ion

4 Design Iteration

4.1 Initial Design Ideas

At the start of our project, we planned for the whole top piece of the game board to be made from one large sheet of wood. Our idea was to laser-engrave the world map directly into that sheet and also cut the holes for all the buttons and LCD screens in the same piece.

We also wanted speakers to be a big part of the game. The plan was to play sound effects when the game started and when a player won. Audio was supposed to be one of the main features.

For the joystick, our original plan was to use the joystick cap that came with the part. We also planned to include more questions per round and make the game a little more complex, with more randomization and longer rounds.

We also expected to build the physical design and electrical system first and save most of the coding for later, because we thought wiring and assembly would take the most time.



Figure 1: Comparison of galvanized metal (right) to weldable steel (left)

Prototyping and Final Design As we started building, a lot of our original plans changed. First, we tested a few different board sizes to make sure the screens, buttons, and joystick fit well. After trying different measurements, we chose a size of about 21" × 24" × 3" because it gave us good spacing without being too big.

The large laser cutter was down, so we couldn't engrave and cut everything from one big sheet like we planned. Instead, we engraved the map on a separate piece of wood and built the board in layers.

We also had to remove the speakers from our final design because the amplifier and wires caused too many issues with our timeline. The joystick cap that came

with the part was too small, so we 3D-printed a taller and easier-to-use one.

Another change was adding a power switch. We didn't plan for one at first, but since we used batteries, we needed a way to turn the game on and off to save battery life.

We also simplified the game rules. Our randomizing code had problems, so we changed the game to 5 questions per player, one joystick choice per turn, and a countdown on the screens. Coding took much longer than we expected, so we learned that software should be started earlier in the process.

4.2 Risk Mitigation During Design

During development, we encountered several challenges that required adjustments to our original plans. To handle these risks effectively, we focused on troubleshooting, iterative design changes, and careful organization of both hardware and software. Some of the key risk mitigations included:

- **Laser cutter unavailability:** We layered a separate engraved piece onto the board instead of cutting the top from a single sheet.
- **Lost or damaged materials:** A previously designed top wood piece was stolen, so we remeasured and rebuilt using spare stock.
- **Electrical component failures:** Two small LCD screens failed and were replaced, while the Arduino GIGA screen required troubleshooting before functioning.
- **Software difficulties:** Coding proved more challenging than expected, so we held regular consultations with ITL helpers to improve our understanding.
- **Wiring and soldering issues:** We reorganized messy wires using wire nuts, redid solder connections carefully, and verified voltages to prevent burned-out LEDs.
- **Joystick adjustments:** The 3D-printed cap restricted movement initially, so we trimmed and secured it with super glue for proper operation.

These mitigations allowed us to adapt our design efficiently, ensuring that the final build was both functional and user-friendly while staying on schedule.

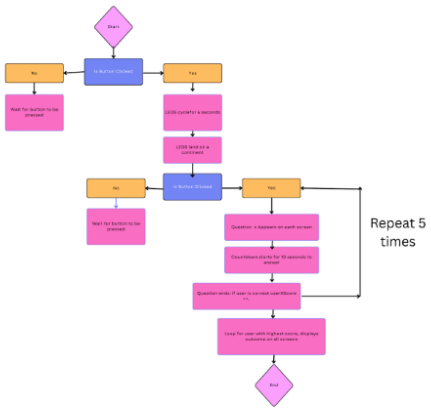


Figure 2: Block Diagram

5 Electrical Design

Our electrical design is based around the Arduino GIGA we choose for the amount of pins, which acts as the main controller for the whole game. All of our components connect back to the GIGA, which includes the LED's, push buttons, joystick, and smaller LCD screens. Each part of the system needs a steady 5V, so we powered everything with a 7.4V rechargeable lithium-ion battery, which the GIGA regulates down to needed 5V.

5.1 Block Diagram

5.2 Circuits, Components, and Electrical Functions

1. LCD Screens: One large LCD for displaying questions/answers. Four 2x16 LCD for each player's score count. These screens connect to the GIGA through their backpack boards using simple wiring for power and data.

2. Push Buttons: Used 16 buttons, 4 per player (A, B, C, D). When the player presses down, the GIGA reads the input signal and stores their answers. Keeping one side of every button to ground made wiring easier.

3. Joystick: Connected to the GIGA through analog pins. It controls which continent the player selects by reading the closest X and Y values to match the joysticks movement.

4. Power System: Powered by a 7.4V lithium-ion battery, which feeds into

the Arduino GIGA. The GIGA then regulates this 7.4V to a steady 5V across all components.

5. Wiring: Wires ran from each component through the inside of the board and soldered the connections for the LCD screens and buttons directly onto the GIGA shield. Soldering these components gave us reliable connections that wouldn't loosen while the game was being used.

6 Software Design

The software for *World Trivia* serves as the central controller that coordinates user input, game logic, scoring, timing, and communication between all electrical subsystems. The code integrates multiple libraries, handles real-time joystick input, updates three LCD screens, controls LED strips, and manages the flow of the trivia game.

6.1 Code Flow Charts – Describes the logic structure of the game, including labeled, referenced flowcharts.

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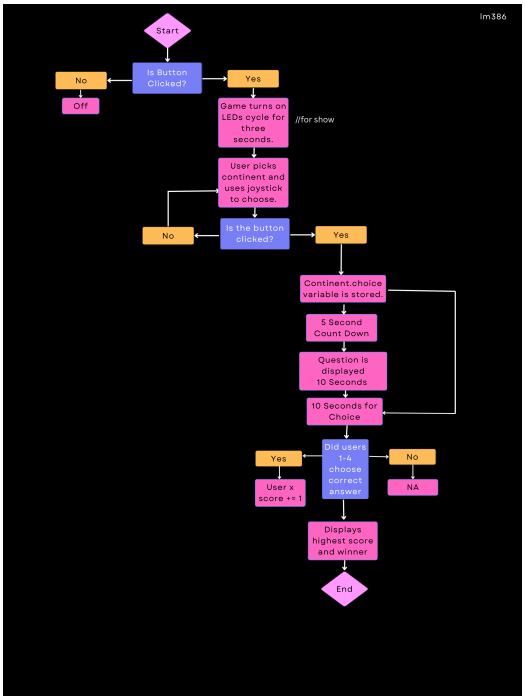


Figure 3: Software flow chart

6.1 Code Flow Charts

Flow charts were created to visualize the overall structure of the code, helping us plan each stage of gameplay and debug issues during development. These diagrams provide a detailed look at how the joystick, LEDs, displays, and scoring system interact.

6.1.1 6.1.1 Main Game Flow

The main game loop begins in **Continent Selection Mode**, where the joystick controls navigation across the world map printed on the board. As the joystick moves, the software reads X and Y analog values, computes the angle, and identifies the continent being pointed to. The LED for that continent lights up, giving immediate visual feedback. When the player presses the confirm button, the game locks in that continent and begins a five-question round.

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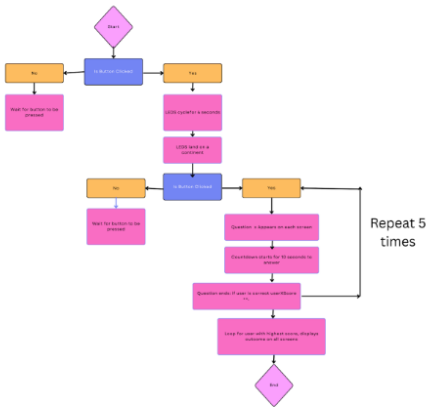


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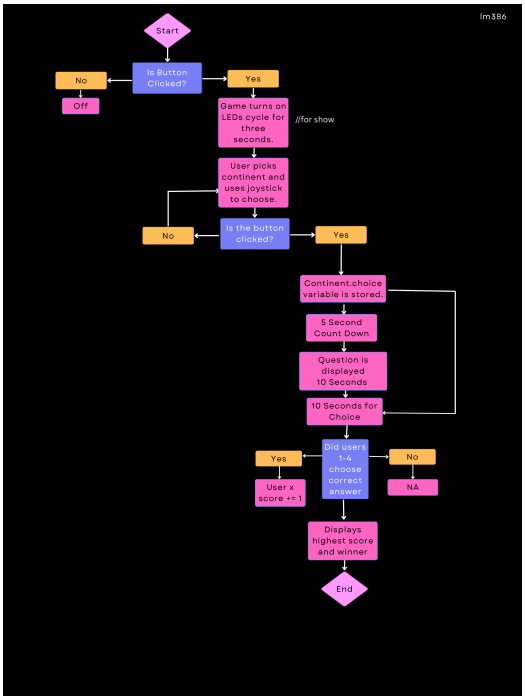


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These diagrams represent how each function leads to the next, helping to understand our structure.

We reference Fig. 7 throughout Section 6 whenever explaining how different subsystems interact.

10.1.2 6.1.2 Joystick Direction Flow

Figure 7: Joystick Input Processing Flowchart

(Insert flowchart)

Description:

This flowchart shows the conversion of raw X/Y analog readings into cardinal directions. Because the joystick is an analog device, we used thresholds and Pythagorean distance calculations to reliably determine whether the player was selecting North, South, East, or West. The diagram highlights:

- Reading `analogRead(xPin)` and `analogRead(yPin)`
- Center dead-zone filtering
- Computing magnitude
- Computing angle using `atan2(y, x)`
- Mapping angle ranges to one of four valid outputs

This figure is referenced in Section 6.3 when discussing programming issues.

10.2 Block Diagram

10.3 Programming issues

Pythagorean Theorem for joystick, keeping game score More details How much coding Libraries necessary language used

10.4 Code Location

Our code is located in a Madison's saved files.

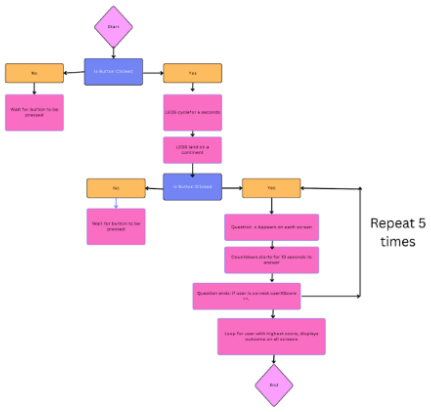


Figure 7: Block Diagram

11 Mechanical Design

World Trivia is primarily driven by electronics and software, but the mechanical elements played a key role in making the game usable, durable, and engaging for players. Our mechanical design centered on creating a system that would be easy for elderly users to interact with while also maintaining a clean, polished appearance. The major physical components include a custom 3D-printed joystick topper, a laser-engraved world map panel, and a hand-built wooden enclosure designed to house the LEDs, LCD screens, Arduino GIGA, and user input buttons.

11.1 Mechanical Components, Fabrication, and Assembly

To make continent navigation intuitive, we designed a custom compass-rose-style joystick topper that clips securely onto the purchased joystick. It took two iterations to get the fit and height correct; after printing, we used the lathe to add additional clearance so the joystick could reach all required angles for continent selection. The design was created in OnShape, allowing us to easily adjust tolerances and geometry between versions.

The top of the board features a laser-engraved map of the world, which visually supports the gameplay and gives players a clear reference for which continent they are selecting. Due to equipment limitations, we engraved the map on a smaller operational laser cutter, then cut the full panel separately once a larger laser cutter became available. The engraved top panel sits above the electronics,

with LEDs positioned behind continent locations to light up during selection.

The enclosure itself was built from stock wood. We cut the material to size using the table saw, band saw, and miter saw, and then refined the interior cavities using the drill press and mill to create precise slots for components. The final enclosure dimensions are 24 inches wide by 21 inches long, and 3 inches tall, providing enough internal space for wiring, the Arduino GIGA, four LCD score screens, and all input hardware.

11.1.1 Tools

Throughout this project we used the 3D Printer, Laser Cutter, Lathe, Table Saw, Band Saw, Miter Saw, Drill Press, Mill Saw, Hand Files, Soldering Irons, Wire Tip Tinner, and Wire Crimper.

11.1.2 7.1.1 Assembly of the Enclosure

The wooden box was assembled using wood glue, air-gun tacks, and screws to ensure structural stability. Electrical components—including the LCD score screens, Arduino GIGA, joystick, and interior wiring—were mounted using screws for durability and accessibility. The answer buttons required additional stability, so they were secured with wooden shims. LEDs were fixed in place with adhesive to align them accurately with the engraved continents. A removable back panel was attached with screws to allow access for maintenance and future modifications.

12 Testing and Analysis

This should be a large section with graphs and tables as appropriate. We will insert all charts of data and testing photos.

12.1 Electrical Testing

Our electrical components required extensive testing. Before connecting them to our Arduino, we tested them for manufacturing flaws to ensure they were safe to use. We then tested each element with simple code on an Ardunio Uno to ensure they ran correctly. In the connection process, we ran into some issues with soldering due to our electrical schematic, which we might layout differently if we had the opportunity for a new iteration. This schematic had our power, input and ground wires all connected in a tight bundle on the shield. This made it both difficult to trace wires and easy to create a short in the wiring. After finding these wiring shorts, we repaired them. We then began ensuring each component had no shorts, as we learned our original attempt of pin to pin soldering was an inferior method to other connections, such as wire nuts and crimped wire joints. We swapped out connections with potential for shortage for these other methods of connection. Two of the LCD screens were not working properly after testing because of damage along our wiring process. These were rebuilt using new LCD backpacks and were then treated with more delicacy. The cut length of LEDs were installed and intially tested with code delivering too high of voltage for their rating. Those LEDs burned out and replaced with our excess LEDs which we used with less voltage.

12.1.1 Qualitative and Quantitative Analysis

For our quantitative testing, we played the game through with several users, tracking how long it took to play the game, how long it took for a player to figure out how to play the game, the average time it took a player to answer a question, and the response time of the game to an input.

For our qualitative testing, we asked users about their experience with their game and for any potential feedback they had. Users reported the game was fun but challenging, the joystick was fun to use, and they liked the laser engraved world map. We also observed players reactions to the game, many leaned in, excited to touch the joystick and see the LEDs light up. Our team was excited to see other people excited about our game.

12.2 Software Testing

We will expand this to be a summary of our testing in section 6.

12.3 Overall Results of Testing

13 Timeline and Bill of Materials

13.1 Gantt Chart

This section outlines the overall project timeline and the resources required to build World Trivia. The schedule includes major milestones such as design, prototyping, testing, and final integration. The Bill of Materials (BOM) summarizes all components used in the project along with their quantities and approximate costs. Figure 4 shows the project timeline from initial concept development through the final Expo presentation. Major phases include:

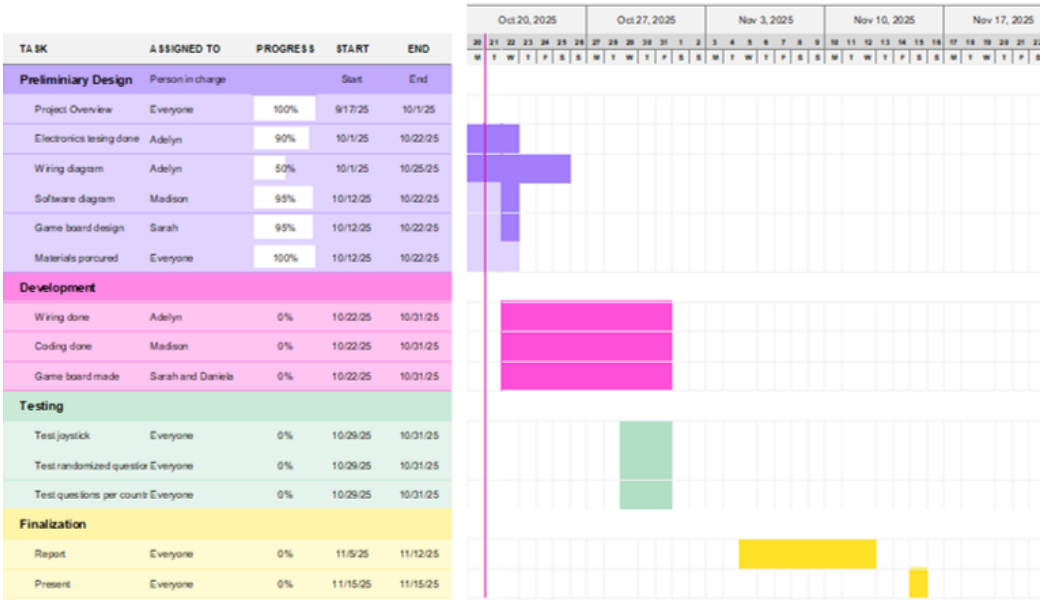
- **Brainstorming and Design**
- **Initial Testing**(component testing)
- **Material purchasing**
- **Wiring and Coding Prototyping**
- **Mechanical Build** (3D printing, laser engraving, enclosure construction)
- **Integration Testing**
- **Qualitative and Quantitative Testing**

The Gantt chart provides a week-by-week overview of how the team divided work, overlapped tasks, and tracked progress to ensure the project met the final deadline.

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13.2 Bill of Materials

The Bill of Materials lists every electrical, mechanical, and structural component required to build the game. This includes the Arduino GIGA, LCD screens, LED strips, joystick, buttons, power components, wiring, wood stock, fasteners, and 3D printing or laser-cut materials. width=12025-11-20 202548.png



TASK	ASSIGNED TO	PROGRESS	START	END
Preliminary Design	Person in charge		Start	End
Project Overview	Everyone	100%	9/17/25	10/1/25
Electronics tesing done	Adelyn	90%	10/1/25	10/22/25
Wiring diagram	Adelyn	50%	10/1/25	10/25/25
Software diagram	Madison	95%	10/12/25	10/22/25
Game board design	Sarah	95%	10/12/25	10/22/25
Materials porcured	Everyone	100%	10/12/25	10/22/25
Development				
Wiring done	Adelyn	0%	10/22/25	10/31/25
Coding done	Madison	0%	10/22/25	10/31/25
Game board made	Sarah and Daniela	0%	10/22/25	10/31/25
Testing				
Test joystick	Everyone	0%	10/29/25	10/31/25
Test randomized questior	Everyone	0%	10/29/25	10/31/25
Test questions per count	Everyone	0%	10/29/25	10/31/25
Finalization				
Report	Everyone	0%	11/5/25	11/12/25
Present	Everyone	0%	11/15/25	11/15/25

Figure 8: Gantt Chart

14 Conclusion

With World Trivia's ease of use, design functionality, and client appeal, it would be a successful product on the market. We plan to expand our conclusion to include the positives from testing to make a more complete argument.

World Trivia successfully combines electronics, software, and mechanical design to create an engaging, multi-player educational game. By focusing on intuitive controls, clear visual feedback, and real-time scoring, the project delivers a game that is both fun and accessible, especially for our intended audience of elderly players. The combination of joystick-based continent selection, LED feedback, the Arduino GIGA display for questions, and individual LCDs for scoring created a smooth and enjoyable gameplay experience.

14.1 10.1 Summary

14.1.1 10.1.1 Technical Summary

From a technical perspective, the project demonstrates effective integration of multiple hardware components and software modules. The joystick, LED sys-

Quantity & Product	Price per
Arduino GIGA with LCD Screen	$121.00
2x16 LCD Screens	$15.00
RGB LEDs	$16.00
Joystick	$7.00
Speaker Wires	$6.00
Acrylic Sheet	$33.00
Wood	$30.00
Power Supply	$29.00
Total	$257.00

Figure 9: Bill of Materials

tem, GIGA display, answer buttons, and LCDs all work together reliably. Our code architecture, using modular design and a state-machine approach, ensures responsive input handling and accurate score tracking, even when multiple players interact simultaneously.

14.1.2 10.1.2 Design Takeaways

Mechanically, the combination of 3D-printed joystick components, laser-engraved maps, and a carefully designed wooden enclosure resulted in a game that is both functional and visually appealing. Iterative testing and adjustments during fabrication improved usability and durability. The placement of each component was designed to maximize accessibility and player comfort.

14.1.3 10.1.3 Teamwork Takeaways

This project also reinforced the importance of communication, planning, and collaboration within a team setting. Coordinating electronics, software, and mechanical design required frequent check-ins and problem-solving as a group. Through these experiences, we learned to divide tasks efficiently while maintaining a shared understanding of project goals.

14.2 10.2 Future Work

While World Trivia meets its design goals, there are opportunities for future improvements:

- **Expand question database**: Adding more questions or categories would increase replay value. We could create a function to switch between genres of trivia within continents, such as a sports trivia of the continents game, in addition to random continent trivia.
- **Enhanced visual feedback**: Speakers to add audio feedback could open the project to new audiences. A speaker reading off the questions could allow those with visual impairments to participate in the game. Speakers could also be used for enhancements like a countdown song or sound effects. We could also add coding for animations that could make the game even more exciting.

- **Adaptive difficulty**: We could create a system to adjust question difficulty based on player performance.
- **Portability and durability**: Refining the enclosure for lighter weight or modular assembly could make the game easier to transport.

These improvements would further enhance the educational value, usability, and overall enjoyment of the game.

14.3 Summary

World Trivia successfully demonstrates the integration of electronics, software, and mechanical design into a fully functional, multi-player educational game. The project achieves its primary goals: an intuitive joystick interface for selecting continents, LED feedback for continent selection, a GIGA screen for trivia questions, and individual LCDs to display player scores.

14.3.1 Technical Summary

The software and hardware systems worked together reliably. The joystick accurately mapped to continents using angle calculations, the LEDs responded instantly, the Arduino GIGA displayed questions clearly, and each player's LCD updated scores in real time. The modular code structure and state-machine design allowed smooth operation and minimized conflicts between inputs and outputs.

14.3.2 Design Takeaways

Mechanically, the combination of a 3D-printed joystick topper, laser-engraved map, and custom wooden enclosure created a system that was easy to use, durable, and visually appealing. Component placement was carefully planned to maximize accessibility and ensure a positive user experience.

14.3.3 Teamwork Takeaways

The project highlighted the importance of collaboration and communication. Coordinating electrical, software, and mechanical tasks required regular team meetings, shared documentation, and iterative problem solving. We all learned the importance of these skills and how critical they are to completing a project successfully and efficiently.

14.4 Future Work

15 Appendix